SABRINE BEJAOUI

Second-Year Professional Master's Student in Decision Support Information Systems and Data Science

Data Science | Business Intelligence | Generative AI | Final-year internship from February 2027

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PROFILE

Second-year Master's student in Decision Support Systems & Data Science, seeking a 6-month end-of-studies internship in Data Science, Machine Learning, or Artificial Intelligence. Motivated by solving real-world challenges, I aim to apply my technical and analytical skills to innovative projects that deliver meaningful business value.

EDUCATION

Professional Master's Degree in Decision Support Information Systems and Data Science

Higher Institute of Management of Bizerte (ISG Bizerte), Tunisia | 2025 – 2027

Bachelor's Degree in Business Computing, specialization in Business Intelligence

Higher Institute of Management of Bizerte (ISG Bizerte), Tunisia | 2022 – 2025

PROFESSIONAL EXPERIENCE

ERP and Sales Administration Assistant

Société Sanitaire, Bizerte, Tunisia | Summer job | Jul 2025 – Sep 2025

- Managed and used the company's ERP (Enterprise Resource Planning) system to maintain customer, supplier and contact information.
- Prepared quotes, orders and invoices in the ERP, and monitored customer payments and outstanding receivables.
- Handled cash register operations and day-to-day administrative tasks.
- Produced sales, customer and payment reports, and checked consistency between sales, inventory and invoices.

Final-Year Internship

LEONI Wiring Systems, Bizerte, Tunisia | Feb 2025 – May 2025

- Designed and developed a digital solution for real-time monitoring of production-line productivity, replacing paper-based reporting.
- Centralized production data from several sources and improved its reliability through ETL pipelines and a data warehouse.
- Developed a web application allowing teams to record hourly output and headcount.
- Designed dashboards and alerts to quickly identify waste and performance deviations, supporting the Lean approach.
- **Technologies:** Python, FastAPI, PostgreSQL, React, Node.js

ACADEMIC PROJECTS

IntelliHire – Intelligent Recruitment Assistant | NLP and Machine Learning

- Developed an application that matches PDF resumes against job offers and ranks candidates by relevance.
- Extracted resume text (PyMuPDF, OCR) and identified skills using named entity recognition (spaCy).
- Applied lexical (TF-IDF, BM25) and semantic similarity measures (Sentence Transformers, cosine similarity) to show matched, partial and missing skills.
- **Technologies:** Python, spaCy, scikit-learn, Sentence Transformers, FastAPI, PostgreSQL, React, TypeScript

FraudLens | Financial Fraud Detection and Investigation Platform

- Built a multi-organization financial fraud detection platform that turns transaction files into prioritized, explained alerts for analyst investigation.
- Trained and compared Random Forest, XGBoost, LightGBM, Logistic Regression and Isolation Forest on a highly imbalanced public dataset, evaluated with PR-AUC, precision and recall on a temporal split.
- Explained each alert with SHAP and added data validation, model drift monitoring and an audit trail to support reliable decisions.
- **Technologies:** Python, scikit-learn, XGBoost, LightGBM, SHAP, FastAPI, PostgreSQL, React, TypeScript

TECHNICAL SKILLS

Programming and Data Analysis: Python, R, SQL, Pandas, NumPy, Scikit-learn, Statsmodels
Artificial Intelligence and Machine Learning: Machine Learning, Deep Learning, NLP, Generative AI, TensorFlow, XGBoost, LightGBM, SHAP, spaCy, Transformers, Sentence Transformers, LangChain, RAG, MLflow
Business Intelligence and Data Visualization: Power BI, SSIS, Talend, Streamlit
Databases: Relational (MySQL, PostgreSQL, SQL Server); NoSQL (MongoDB, Redis, Apache Cassandra, Neo4j)
Web Development and Frameworks: React, TypeScript, Node.js, Express.js, Flask, FastAPI, REST APIs
Development Tools: Git, GitHub, Docker

LANGUAGES

Arabic: Native | French: Professional proficiency | English: Professional proficiency